

# Boundary Residue Notes

## Repunit Inverse Towers, Brun Deficit, and a Collatz Bay

Conversation wrap / editable working note

5 June 2026

### Abstract

This note wraps the chat into a modifiable mathematical sketch. The central observation is not a PSLQ-style identity. It is a shared boundary-residue pattern: an upper branch of an inverse repunit tower produces a residue just above 1, while Brun's constant and the proposed Collatz Bay snap-back constant sit just below 2. The close numerical match is strongest between the tower residue  $z_+(1.11) - 1$  and the Brun deficit  $2 - B_2$ .

## 1 Status

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|                   |                                                                                                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Keep</b>       | The repunit iteration has a clean attracting fixed point branch, and its upper residue can be defined exactly as a function of $c$ .                                  |
| <b>Caveat</b>     | The closeness to Brun's constant is numerically real but tunable; it is not evidence of an identity by itself.                                                        |
| <b>Quarantine</b> | Any claim that Brun's constant, the Collatz Bay constant, and the repunit tower are the same hidden object.                                                           |
| <b>Next test</b>  | Treat the pattern as a boundary-shape diagnostic: compare residual profiles, derivatives, stability exponents, and cycle structure across families of iterative maps. |

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## 2 The code fragment

The numerical experiment from the chat was:

```
c = 1.11

y = 1.0
for _ in range(1000):
    y = c ** (- (c ** y))
print("Lower_bound_(starts_with_neg):", y)

z = 1.0
for _ in range(1000):
    z = c ** (c ** (-z))
print("Upper_bound_(starts_with_pos):", z)
```

With  $c = 1.11$ , this converges numerically to

$$y_- \approx 0.8917774506448141,$$
$$z_+ \approx 1.0975340696153890.$$

The upper branch is the important branch for the Brun comparison because

$$z_+ - 1 \approx 0.0975340696153890.$$

### 3 Upper inverse-tower branch

The upper iteration is the dynamical system

$$T_c(z) = c^{c^{-z}}.$$

An attracting fixed point satisfies

$$z_+(c) = c^{c^{-z_+(c)}}.$$

Define the tower boundary residue by

$$L_{\text{tower}}(c) = z_+(c) - 1.$$

For  $c = 1.11$ ,

$$L_{\text{tower}}(1.11) \approx 0.0975340696153890.$$

Let  $L = \log c$ . Since  $c = 1.11$  is close to 1,  $L$  is small:

$$L = \log(1.11) \approx 0.1043600153242429.$$

The fixed point equation can be rewritten as

$$z = \exp\left(Le^{-Lz}\right).$$

The small- $L$  expansion of the attracting branch is

$$z_+(c) = 1 + L - \frac{L^2}{2} - \frac{4L^3}{3} + \frac{7L^4}{8} + \frac{14L^5}{5} + O(L^6).$$

Therefore the residue  $z_+(c) - 1$  naturally lands near, but below,  $\log(1.11) \approx 0.10436$ .

### 4 Brun deficit

Let  $B_2$  denote Brun's constant,

$$B_2 = \sum_{(p,p+2)} \left( \frac{1}{p} + \frac{1}{p+2} \right),$$

where the sum is over twin prime pairs. A commonly quoted numerical estimate is

$$B_2 \approx 1.902160583104.$$

This gives the Brun boundary deficit

$$L_{\text{Brun}} = 2 - B_2 \approx 0.0978394168960000.$$

The key comparison is therefore

$$L_{\text{tower}}(1.11) - L_{\text{Brun}} \approx -0.0003053472806109,$$

with relative gap

$$\frac{|L_{\text{tower}}(1.11) - L_{\text{Brun}}|}{L_{\text{Brun}}} \approx 0.003121.$$

So the close-looking match is real at naked-eye scale:

$$z_+(1.11) - 1 \approx 2 - B_2.$$

But it should be treated as a diagnostic coincidence until a structural bridge is found.

## 5 Collatz Bay input from the chat

The chat supplied a proposed Collatz Bay snap-back constant

$$c^* \approx 1.9086708647584145, \quad C(c^*) = \frac{\log 2}{\log c^*} \approx 1.07230747.$$

The corresponding deficit from the boundary anchor 2 is

$$L_{\text{Collatz}} = 2 - c^* \approx 0.0913291352415855.$$

This is less close to the tower residue than the Brun deficit is, but it belongs to the same broad residual class near 0.1:

$$L_{\text{tower}}(1.11) \approx 0.09753, \quad L_{\text{Brun}} \approx 0.09784, \quad L_{\text{Collatz}} \approx 0.09133.$$

## 6 Numerical table

| Object               | Definition                     | Value              |
|----------------------|--------------------------------|--------------------|
| Lower repunit branch | $y = c^{-c^y}$ at $c = 1.11$   | 0.8917774506448141 |
| Upper repunit branch | $z = c^{c^{-z}}$ at $c = 1.11$ | 1.0975340696153890 |
| Tower residue        | $z_+(1.11) - 1$                | 0.0975340696153890 |
| Brun estimate        | $B_2$                          | 1.9021605831040000 |
| Brun deficit         | $2 - B_2$                      | 0.0978394168960000 |
| Collatz Bay constant | $c^*$                          | 1.9086708647584145 |
| Collatz deficit      | $2 - c^*$                      | 0.0913291352415855 |
| Reciprocal Brun      | $1/B_2$                        | 0.5257179698089272 |
| Reciprocal Collatz   | $1/c^*$                        | 0.5239247994318668 |

## 7 Boundary residue function

A clean general object is not a classical number-theoretic  $L$ -function yet. It is better described as an  $L$ -like Lyapunov or boundary-residue function.

Let  $M_c$  be an iterative system with an attracting object  $A_c$  and a boundary anchor  $b$ . The boundary residue is

$$\mathcal{L}_\partial(M_c; b) = \text{dist}(A_c, b).$$

Examples:

$$\begin{aligned} \mathcal{L}_\partial(T_c; 1) &= z_+(c) - 1, \\ \mathcal{L}_\partial(\text{Brun}; 2) &= 2 - B_2, \\ \mathcal{L}_\partial(\text{Collatz Bay}; 2) &= 2 - c^*. \end{aligned}$$

This turns the intuition into a modifiable framework:

$$\boxed{\text{compare systems by their stable boundary residues, not by forcing identities.}}$$

## 8 Why the closeness happens

The repunit tower branch is close because  $c = 1.11$  makes  $\log(c)$  small. The map

$$z \mapsto c^{c^{-z}}$$

acts almost like the identity near 1, then pulls the fixed point to

$$z_+ \approx 1 + 0.0975.$$

Brun's constant, by contrast, is near

$$B_2 \approx 2 - 0.0978.$$

So the apparent match is really a residual match:

$$\text{upper tower excess above 1} \approx \text{Brun deficit below 2}.$$

The Collatz Bay constant is similar in shape but looser:

$$c^* \approx 2 - 0.0913.$$

## 9 Tunability warning

Because  $L_{\text{tower}}(c)$  varies continuously with  $c$ , a nearby base can be chosen to match the Brun deficit almost exactly. Numerically,

$$c \approx 1.1103954628296591$$

solves

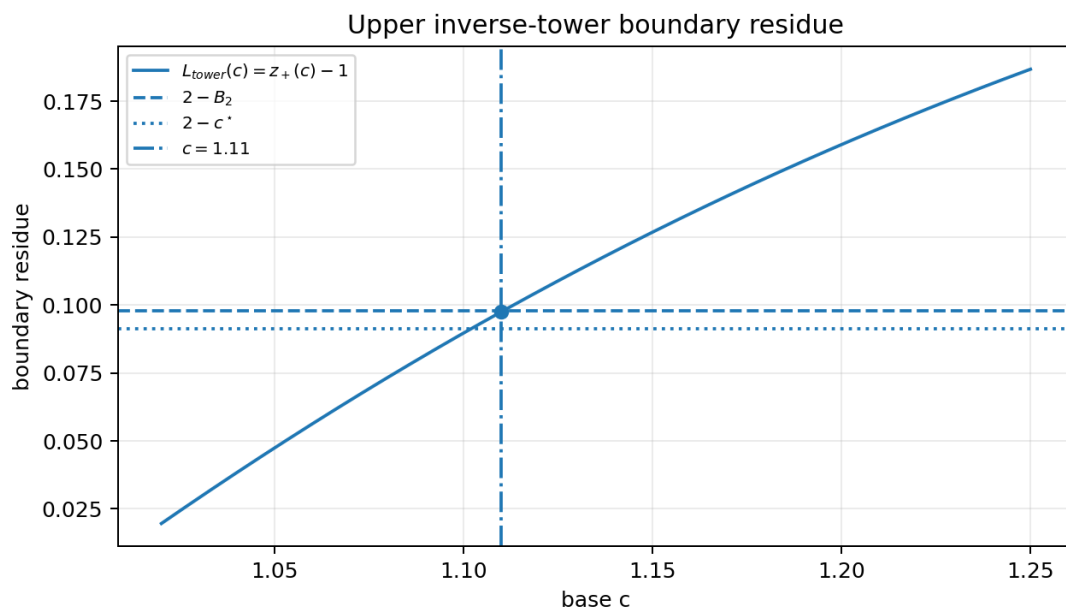
$$L_{\text{tower}}(c) = 2 - B_2.$$

The local sensitivity near  $c = 1.11$  is approximately

$$\left. \frac{d}{dc} L_{\text{tower}}(c) \right|_{c=1.11} \approx 0.772531.$$

This is the main reason not to overclaim. The closeness is useful as a detector, not a proof.

## 10 Shape plot



The plot shows that  $L_{\text{tower}}(c)$  crosses the Brun-deficit level near  $c \approx 1.110395$ . This confirms the match is part of a smooth boundary profile rather than an isolated identity.

## 11 Research program

A safe next development path:

1. Define a library of maps  $M_c$  with attracting fixed points or cycles.
2. Compute  $\mathcal{L}_\partial(M_c; b)$  and its derivative profile.
3. Compare boundary residues only after normalizing the boundary anchor  $b$ .
4. Add stability data: multiplier, Lyapunov exponent, basin size, and period of the attracting cycle.
5. Only then test whether any arithmetic constants land on non-tunable special points of the profile.

## 12 Minimal conclusion

The plausible object is

a boundary-residue L-profile for iterative systems.

It is not yet a Dirichlet-style  $L$ -function. The honest claim is:

$L_{\text{tower}}(1.11) \approx L_{\text{Brun}}, \quad L_{\text{Collatz}}$  is nearby but weaker.

That is enough to preserve the idea without freezing it into a false theorem.

## References

- [1] E. W. Weisstein, *Brun's Constant*, MathWorld—A Wolfram Web Resource. <https://mathworld.wolfram.com/BrunsConstant.html>
- [2] D. Platt and T. Trudgian, *Improved bounds on Brun's constant*, arXiv:1803.01925, 2018. <https://arxiv.org/abs/1803.01925>
- [3] P. Sebah and P. Demichel, *Introduction to twin primes and Brun's constant computation*, 2002. <https://numbers.computation.free.fr/Constants/Primes/twin.html>